

WATERWATCH · REQUIREMENTS

WaterWatch requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	WWT-REQ-001	Version	0.3
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwatch		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	For TRL 3, status of every requirement from WWT-CAL-001; decisions from WWT-DDR-001 recorded; targets unchanged
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

WaterWatch requirements

These requirements were first set for the concept and are now checked by calculation in WWT-CAL-001. Targets are unchanged from v0.2; they are still proposals for review, not user-validated needs, and will be revised after co-design. On paper, two requirements are **not met** (R1 and R10), three are at risk (R2, R9 and R11), five are met by calculation and three by design. Amish's decisions on the TRL 2 review are in WWT-DDR-001.

Table 1. Requirements and TRL 3 status (from WWT-CAL-001, Table 4).

ID	Requirement	Target	Verification	TRL 3 status
R1	Measure free chlorine at the tap	0 to 2.0 mg/L; within ± 0.1 mg/L below 1.0 mg/L and ± 15 % above, at pH 6.5 to 8.5 and 5 to 40 °C	Error budget (WWT-CAL-001, E); later side-by-side with a DPD comparator	Not met: with site pH the error is ± 0.131 mg/L at 0.5 mg/L and pH 7.5 before any sensor drift; only pH 7.0 meets the target. A relaxed target is proposed in docs/REVIEW.md , awaiting Amish
R2	Measure turbidity	0 to 100 NTU, reporting "above 100 NTU" beyond; within ± 0.5 NTU or ± 10 %, whichever is greater	Published open-source design; bubble and settling calculation (WWT-CAL-001, F); later stabilized standards	At risk: the 30 s wait clears 100 μ m bubbles at 5 °C; window fouling between cleans is unknown
R3	Measure water temperature	0 to 50 °C, within ± 0.5 °C	Sensor datasheet	Met by design
R4	Sample on a schedule	One reading per hour by default; configurable from 15 min to 24 h	Energy at the 15 min schedule (WWT-CAL-001, A and B)	Met by design (15 min schedule: 1.589 Wh/day, 9.7 days on battery, but 72 L/day of water)
R5	Alert the people who act	SMS to up to three numbers within 15 min of a confirmed crossing (two consecutive readings): free chlorine below 0.2 mg/L, turbidity above 5 NTU, device fault or low battery; thresholds adjustable per site	Latency calculation (WWT-CAL-001, H)	Met on paper where there is coverage: 13.0 min worst case with three tries at 3 min spacing
R6	Report and keep data	Upload readings at least every 4 h over LTE-M, NB-IoT or 2G (LoRaWAN variant); keep 90 days on board if the link fails; open CSV or JSON format; works without a cloud service	Data budget (WWT-CAL-001, G)	Met on paper: 0.86 MB per month; 90 days in 138 kB
R7	Run on sunlight alone	7 days or more with no sun from full; back to full in 3 clear days or fewer	Energy budget (WWT-CAL-001, A and B)	Met on paper: 28.6 days (24.3 at 0 °C); 1.18 clear days to refill; hot days rely on the sun shield (R9)
R8	Use little treated water	20 L/day or less at hourly sampling	Flow calculation (WWT-CAL-001, D)	Met on paper, thin margin: 18.0 L/day nominal, 19.8 L at the regulator's tolerance
R9	Survive outdoors	Electronics IP65 or better; ambient 0 to 45 °C in direct sun; UV-stable parts; flow cell opaque to daylight; battery charging blocked below 0 °C and above 45 °C	Thermal estimate (WWT-CAL-001, C); parts selection	At risk: with the sun shield the inside peaks at 50.8 °C on a 45 °C day (69.8 °C without); the shield factor is assumed
R10	Be maintainable by a caretaker	One visit per month or less, 30 min or less, no special tools: clean the cell, compare with a DPD kit, adjust calibration from a phone or keypad	Task analysis (WWT-CAL-001, J)	Not met on present evidence: the visit takes 30 min, at the limit, and a monthly interval needs chlorine drift below about 0.04 mg/L per

ID	Requirement	Target	Verification	TRL 3 status
				month, which no verified data show
R11	Install simply and safely	Two people, 2 h or less, hand tools only; one tee and isolation valve on the riser; backflow prevented; wetted parts food-safe	Task analysis (WWT-CAL-001, J); design review	At risk: 140 min critical path with the footing cast on the day (65 min if cast earlier); the design parts are met
R12	Stay within the concept budget	Parts \$300 or less per unit, excluding the shared calibration kit and airtime	Priced BOM (WWT-CAL-001, K)	Met on paper: \$282
R13	Report measurements, not verdicts	Messages state what was measured and when; the device never says the water is "safe"	Design review of alert wording	Met by design

■ Assumptions

- Site: public tapstand on a piped, chlorinated rural scheme, with a supply pressure of 0.5 to 4 bar (7 to 58 psi) at the riser. Tapstands come first; tank outlets and kiosks later (WWT-DDR-001, D8).
- Alert thresholds follow the field guidance summarized in WWT-PRB-001: at least 0.2 mg/L free chlorine at the point of delivery, and chlorination losing effectiveness as turbidity rises. The defaults of 0.2 mg/L and 5 NTU were decided by Amish on 2026-09-25 (WWT-DDR-001, D6) and remain adjustable per site.
- Free chlorine accuracy targets follow the resolution of common DPD comparator kits (about 0.1 mg/L), because the caretaker's DPD test is the field reference. WWT-CAL-001 shows that this makes the target as tight as the reference itself.
- pH affects the split between hypochlorous acid and hypochlorite, and amperometric sensors respond mainly to hypochlorous acid. R1 is assessed with a site pH entered at installation and checked monthly (WWT-DDR-001, D2), assumed to move no more than ± 0.2 between visits; a pH probe is a documented variant.
- Budget: the \$300 in `project.yaml` covers one unit's parts. A shared calibration kit (about \$60) and airtime (about \$1 to \$3 per month) are listed separately; both are estimates.

Safety: Meeting these requirements does not make the water safe to drink or the unit safe to install. The sample line connects to a pressurized drinking water supply and the unit contains a lithium iron phosphate battery; see WWT-PRC-001, Safety.